

somics spatial-omics dataset collection

Distribution of the datasets collected in the somics repository (`data/datasets.csv`), covering spatial-omics technologies, modalities, organisms, tissues, and publication years.

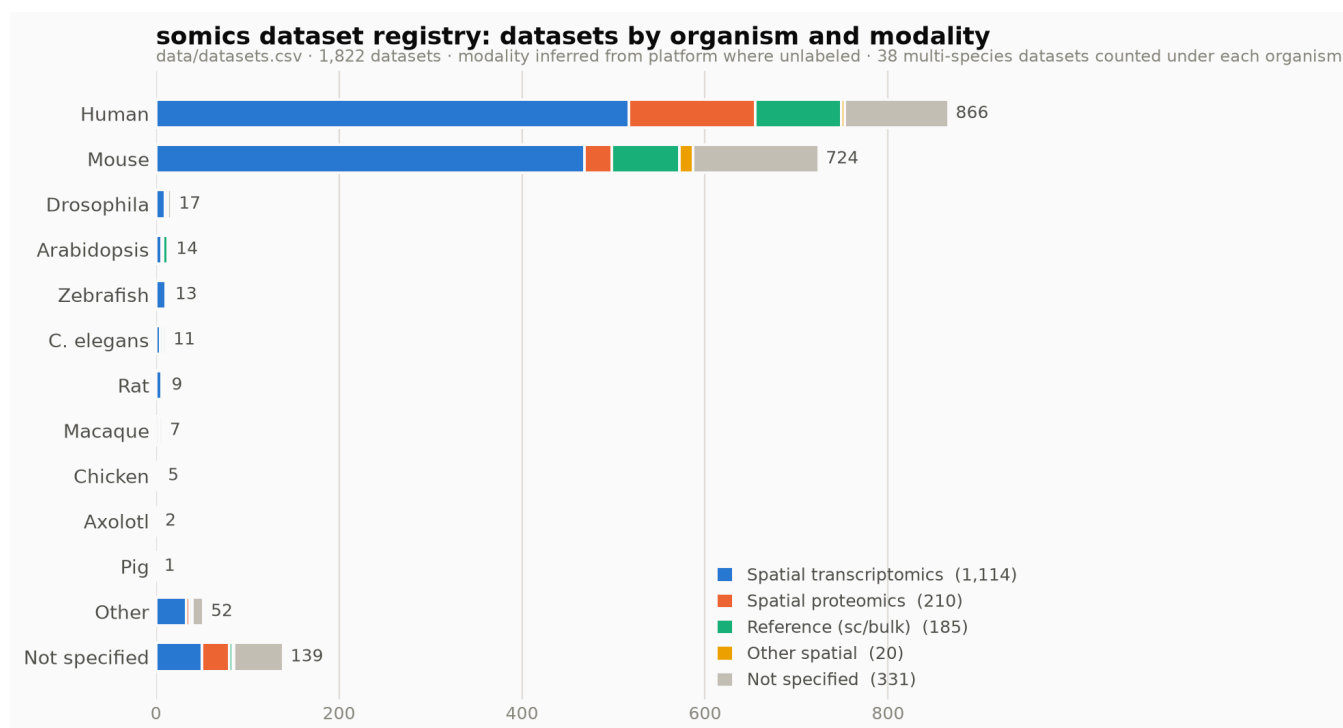
Generated 2026-08-18 from `data/datasets.csv` (1,822 datasets), `data/literature_datasets.csv` (2,429 claim rows from 490 papers) and `data/model_dataset_usage.csv` (3,771 usage rows).

At a glance

Curated datasets	1,822
Source papers mined	490
Datasets with a resolved original publication	1,158 (64%)
Datasets with a data access link	964 (53%)
Datasets with a direct download URL	324 (18%)
Datasets first released by the paper analysing them	651 (36%)
Datasets carrying a perturbation annotation	69
Distinct named models observed	95

The registry is one row per dataset, keyed to the paper that **first released** the data. Where a dataset debuted in a model paper — TERRA's in-house Xenium pancreas, KRONOS's private cohorts — that model paper is its original reference.

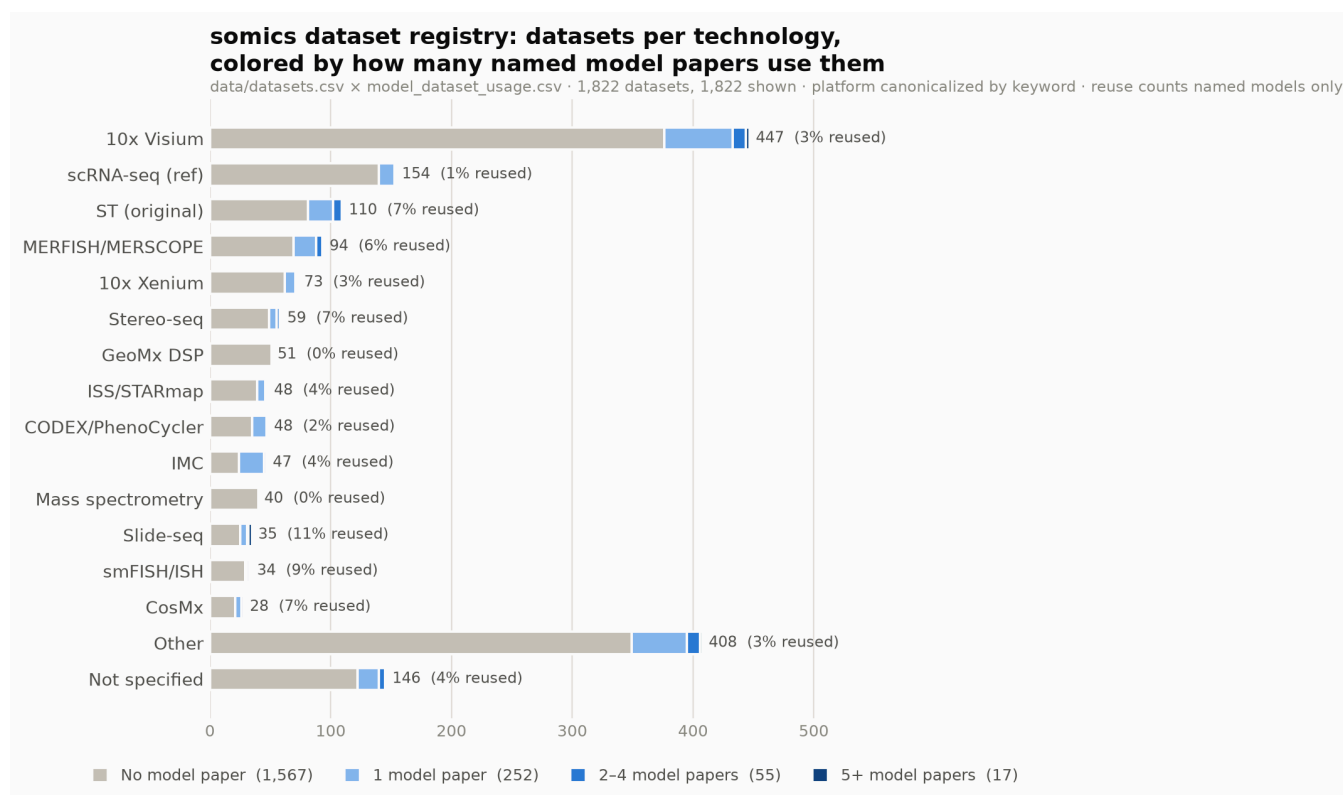
Modality and organism



Spatial transcriptomics dominates at **954 datasets**, against **133** for spatial proteomics; **735** remain unclassified because the platform string didn't match either keyword set, and a blank is preferred to a guess.

Human (**866**) and mouse (**724**) account for almost everything. The long tail is real but thin: Drosophila 17, Arabidopsis 14, zebrafish 13, C. elegans 11, rat 9, macaque 7. Thirty-eight datasets span more than one organism and are counted under each, so the bars sum to slightly more than 1,822.

Technology, and who actually uses it

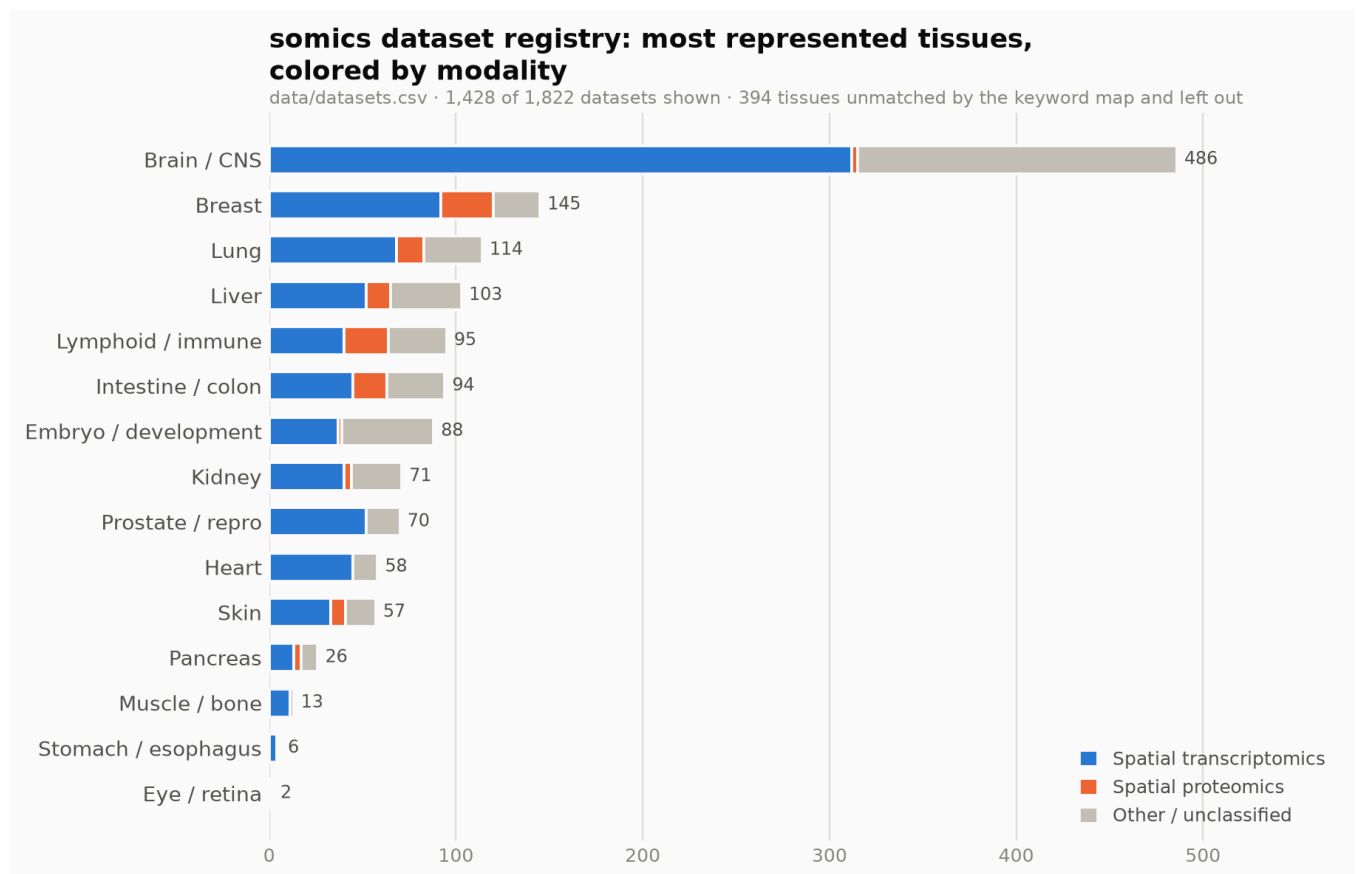


10x Visium is the single largest technology at **447 datasets** — more than the next three combined — followed by MERFISH/MERSCOPE (94), Xenium (73), Stereo-seq (59) and GeoMx DSP (51). On the proteomics side CODEX/PhenoCycler (48), IMC (47) and MIBI trail well behind.

The striking number is the grey band: **1,567 of 1,822 datasets (86%) have no named model using them**. Only 252 are used by one named model, 55 by two to four, and 17 by five or more. Reuse is concentrated in a handful of classics — the DLPFC Visium sections, MERFISH mouse brain, MOSTA — while most of the registry has never been touched by a published model.

Two caveats on that number. Every dataset has at least one row in the usage table by construction, because the paper reporting it becomes a usage row; "used by a model" here means a usage row whose model field carries an actual model name rather than a fallback paper id. And that name is detected from a **Name:** prefix in the paper title, so a model paper titled without one is undercounted.

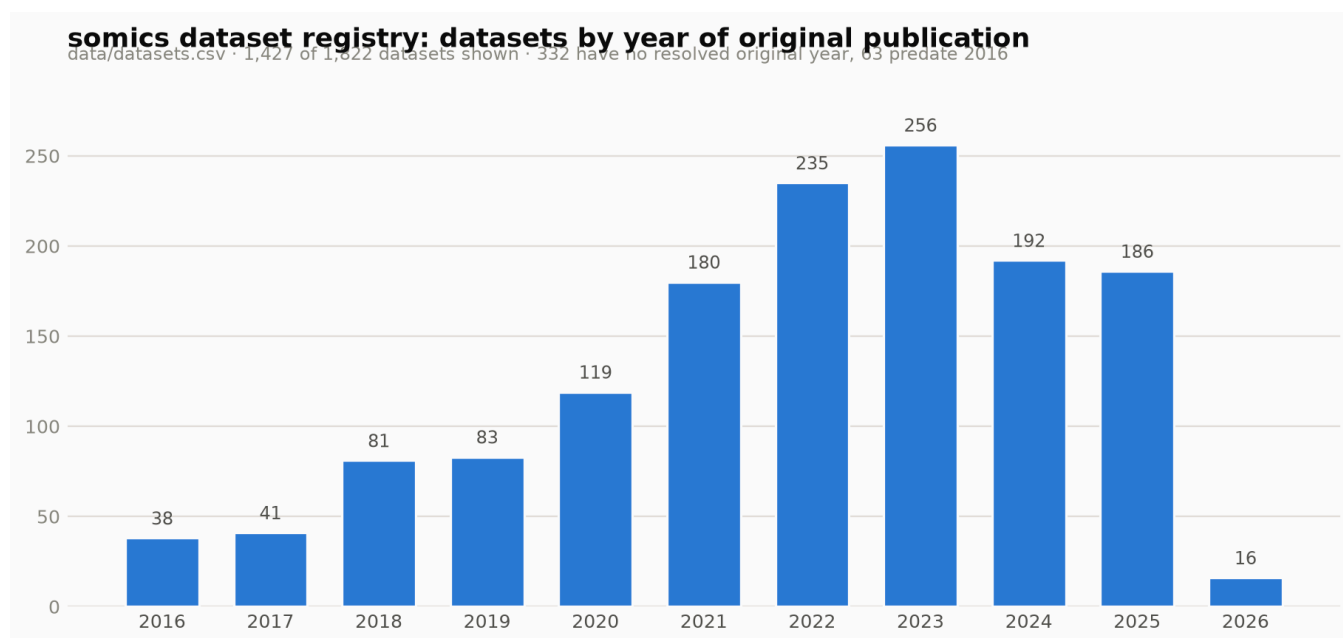
Tissue coverage



Brain and CNS dominate at **486 datasets**, roughly three times the next tissue. Breast (145), lung (114), liver (103), lymphoid/immune (95) and intestine/colon (94) form the second tier. Stomach/esophagus (6) and eye/retina (2) are close to absent — a genuine gap rather than a search artefact, since the organ sweep specifically queried them.

394 datasets are excluded here because their free-text tissue didn't match the keyword map; they are not silently folded into a bucket.

When the data was published



Dated by the *original* publication, the collection tracks the field's growth from the 2016 introduction of spatial transcriptomics through a steep ramp in 2022–2024. Recent years are undercounted, not shrinking: paperclip's bioRxiv ingestion lags publication by roughly three months, so the newest work is systematically missing.

How the collection was built

Four stages, all re-runnable from the repo (see `.claude/skills/harvest-datasets/SKILL.md`):

1. **Search** — paperclip queries over the full-text corpus. Coverage is a function of vocabulary, and each new axis has exposed a large blind spot: a perturbation sweep came back 74% unmined, an atlas/reference sweep 92%, a consortium/organ sweep 46%.
2. **Claim extraction** — one `map` pass per paper into a strict JSON schema, producing `literature_datasets.csv`: one row per (dataset × source paper).
3. **Original-publication tracing** — a second pass expands each paper's dataset citations from its own reference list, deduped by (original publication × platform) and resolved to DOIs through Crossref. This is what makes `datasets.csv` one row per dataset.
4. **Link resolution and verification** — `resolve_download_urls.py` turns accessions into fetchable URLs (Zenodo, GEO, Dryad); `verify_downloads.py` probes each link.

Known limits

- **Link verification is partial.** 1,028 claim rows carry a verdict (572 yes / 66 no / 38 unverified / 352 no link); the 1,401 rows added since that pass are unverified.

- **664 datasets have no resolved original publication** — Crossref couldn't confidently match the citation, or the source is a vendor or portal resource.
- **735 datasets have no modality** and **394 no mapped tissue** — unmatched free text, left blank deliberately.
- **The perturbation column is a partial back-fill** (69 rows) joined from claim rows, not a native field of the trace schema.
- **Extraction is LLM-based**, spot-checked rather than verified row by row. The one exhaustively verified pass was a 38-candidate link-filling round, of which 12 were kept.
- **Dedup is conservative** — same original publication and platform family — so near-duplicates from differently-worded citations may remain.